

UST'-USA KRASNOYARSK, Russian Federation

WMO: 360830

Lat: 52.057N

Lon: 92.139E

Elev: 577

StdP: 94.58

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.8	-29.1	-34.0	0.2	-30.5	-32.3	0.2	-29.0	6.5	-19.6	5.2	-19.5	1.3	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	28.8	17.7	26.9	17.0	25.1	16.5	19.2	26.0	18.2	24.4	17.5	23.2	2.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	13.0	21.4	16.1	12.3	20.4	15.3	11.6	20.0	57.4	25.9	54.2	24.3	51.8	23.4	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.1	5.0	-32.3	32.5	2.8	2.3	-34.3	34.1	-36.0	35.5	-37.5	36.7	-39.5	38.4		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual														
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-22.4	-16.8	-4.1	4.9	11.1	17.7	19.1	17.3	11.0	3.5	-8.2	-18.4	(6)
(7)		DBStd	14.90	4.08	4.86	5.33	3.99	4.24	3.79	2.43	2.68	3.44	3.89	5.93	5.16	(7)
(8)		HDD10.0	4048	1006	750	437	157	37	1	0	0	30	204	547	880	(8)
(9)		HDD18.3	6314	1264	983	695	402	226	56	19	52	221	461	797	1138	(9)
(10)		CDD10.0	878	0	0	0	6	72	232	283	226	59	1	0	0	(10)
(11)		CDD18.3	103	0	0	0	0	2	37	44	19	1	0	0	0	(11)
(12)		CDH23.3	811	0	0	0	1	63	338	283	122	4	0	0	0	(12)
(13)		CDH26.7	181	0	0	0	0	13	92	57	18	1	0	0	0	(13)
(14)	Wind	WSAvg	1.5	1.2	1.4	1.5	1.7	1.9	1.8	1.7	1.5	1.4	1.3	1.8	1.2	(14)
(15)	Precipitation	PrecAvg	404	12	8	12	21	34	59	83	75	35	24	26	18	(15)
(16)		PrecMax	472	31	16	30	46	62	89	135	119	57	59	50	39	(16)
(17)		PrecMin	329	5	1	2	3	16	25	54	38	14	9	13	4	(17)
(18)		PrecStd	37	5	4	7	10	12	15	20	20	11	11	10	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.3	-2.6	11.9	21.1	28.6	31.9	30.8	28.7	22.5	14.9	5.3	-5.1	(19)
(20)		MCWB	-11.7	-4.4	5.2	10.5	14.4	19.2	18.4	18.2	14.2	9.1	2.6	-6.5	(20)	
(21)		2%	DB	-14.1	-5.5	8.5	17.8	25.0	29.2	28.5	26.3	19.9	12.3	3.3	-7.3	(21)
(22)		MCWB	-14.8	-7.4	3.3	9.1	13.0	17.6	18.4	17.8	14.1	7.4	1.3	-8.2	(22)	
(23)		5%	DB	-16.0	-7.9	6.3	15.1	22.1	27.0	26.6	24.4	18.3	10.4	1.1	-10.0	(23)
(24)		MCWB	-16.6	-9.2	2.0	7.4	12.0	16.5	17.8	16.9	13.2	6.6	-0.9	-10.8	(24)	
(25)		10%	DB	-17.3	-9.8	4.1	12.6	19.5	24.9	24.7	22.5	16.7	8.7	-0.9	-12.0	(25)
(26)		MCWB	-17.8	-10.9	0.5	6.2	10.7	16.1	17.3	16.2	12.1	5.7	-2.7	-12.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.6	-4.3	5.3	11.4	15.7	20.1	20.5	20.1	16.1	9.6	3.0	-6.3	(27)
(28)		MCDWB	-11.3	-2.7	10.9	19.7	26.2	29.9	26.7	25.9	20.5	13.8	4.4	-5.5	(28)	
(29)		2%	WB	-14.8	-7.4	3.7	9.3	13.9	18.7	19.4	18.7	14.9	7.9	1.4	-8.2	(29)
(30)		MCDWB	-14.2	-5.7	8.0	16.4	22.8	26.8	25.8	24.6	19.1	11.5	3.3	-7.3	(30)	
(31)		5%	WB	-16.5	-9.2	2.3	8.0	12.6	17.6	18.4	17.7	13.7	6.9	-0.7	-10.7	(31)
(32)		MCDWB	-16.1	-8.0	6.0	14.4	20.8	25.4	24.4	22.8	17.4	10.1	1.0	-10.0	(32)	
(33)		10%	WB	-17.7	-10.9	0.7	6.6	11.3	16.6	17.8	16.8	12.6	5.8	-2.6	-12.6	(33)
(34)		MCDWB	-17.3	-9.8	3.7	11.9	18.5	23.4	23.2	21.2	16.2	8.6	-1.0	-12.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	4.7	7.4	10.3	11.7	12.0	10.9	8.9	9.0	8.3	6.2	4.3	4.0	(35)	
(36)		5% DB	5.5	8.6	11.9	16.7	17.3	14.2	12.6	12.6	10.5	9.4	5.0	5.3	(36)	
(37)		MCWBR	5.1	7.3	7.4	8.9	7.9	5.5	5.2	5.8	5.7	5.6	4.0	4.5	(37)	
(38)		5% WB	5.6	8.4	11.3	15.0	15.7	12.1	10.5	10.2	9.3	8.6	4.6	4.7	(38)	
(39)		MCWBR	5.2	7.2	7.1	8.3	7.7	5.1	4.9	5.4	5.6	5.4	4.0	4.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.289	0.327	0.384	0.436	0.439	0.426	0.424	0.408	0.368	0.357	0.328	0.294	(40)	
(41)		taud	2.041	1.974	1.912	2.029	2.093	2.247	2.292	2.304	2.341	2.219	2.081	2.012	(41)	
(42)		Ebn at Noon	672	751	779	793	814	830	824	817	810	723	611	580	(42)	
(43)		Edh at Noon	80	115	151	153	151	131	123	115	99	90	77	70	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.25	2.32	3.83	4.85	5.55	6.11	5.58	4.80	3.63	2.19	1.19	0.86	(44)	
(45)		RadStd	0.07	0.12	0.18	0.23	0.38	0.48	0.36	0.35	0.32	0.19	0.11	0.06	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page